Exp:6 HAMMING CODE

Aim:

Write a program to implement error detection and corection using Hamming code concept, Make a test run to input data stream and verify error correction feature.

Sender Code:

```
m=int (input ("Enter the number of data bits"))
data = (input ("Enter data bits")
if len(data)! =m or any (bit not in "01" for
                                    bit in data):
        print ("Invalid data!")
        exit ()
r=0
while (2**r) < (m+r+1):
      r+ =1
print ("Number of parity bits :", r)
n=m+r
hamming ['0']* (n+1)
j=0
for i in range (1, n+1):
      if (i & (i-1)) !=0:
          hamming [i] = data [j]
          j+ =1
for p in range (r):
     parity-position = 2**p
     parity =0
```

```
        for i in range (1, n+1):
            if i & parity_~~position~~ posity
                parity ^= int (hamming [i])
        hamming [parity_position] = str (parity)

hamming_code = " "
for i in range (n,0,-1):
    hamming_code += hamming [i]
print ("original data :", data)
print (" Hamming code :", hamming_code)
```

Receiver code:

```
received = input (" Enter received Hamming Code: ")
if any (bit not in "01" for bit in received):
        print (" Invalid Hamming code!")
        exit()
n = len (received)
r=0
while (2**r) <=n:
        r+=1
Print (" Number of parity bits:", r-1)
hamming = ['0'] + list (receive [::-1])
error_position =0.
for p in range (r):
        parity_posity = 2**p.
```

```
parity=0
for i in range (1,n+1)
        if   i in parity-position
                parity ^= int (Hamming [i])

    if parity ! =0:
          error_position + = parity-position

if error_position ==0:
        print (" No error detected")

else
   print ("error detected!")
   print ("error position from right:", error_position)
   if Hamming [error_position] ==`0':
          Hamming [error_position] = `1'
   else
          Hamming [error_position] = '0'
   corrected_code = " "
   for i in range (n,0,-1)
          corrected-code += Hamming [i]
   print(corrected-code).
```

Result:

Error detection & error correction using Hamming code are learned and executed successfully.

Input:

Enter the Text: HELLO

Output

Binary Data: 01 001 00001 0 001 0101 001 0001 0011 0001
00111

Hamming encoded data: 1 0011 0011 100 0 001 0011 0001
001 011 0111 0001111 0010 0 1111

Data Saved.

Result:

Hence the Hamming code is successfully executed.